THE INJECTIVITY LEMMA HOLDS OVER A VALUATION RING: THE MODULE IS FLAT AND FINITELY PRESENTED AND THE MAP IS UNIVERSALLY INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xypic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AT5 (sources/stacks-flat/pinned-original-source.tex line 7158).

Allowed standard prerequisites (editorial).

- The situation and its three assertions. Same chapter, Situation labelled situation-mod-injective, invoked in the statement. Supplies the data A, B, N, M, u and the three properties to be proved.
- Flatness equals torsion freeness over a valuation ring, and finite presentation of flat finite type modules. More on Algebra, Lemmas labelled more-algebra-lemma-valuation-ring-torsion-free-flat and more-algebra-lemma-flat-finite-type-valuation-ring-finite-presentation, cited at proof lines 7163-7172. Identifies flatness with torsion freeness and upgrades flatness of N/T to finite presentation over B.
- The criterion for universal injectivity and the reduction of test modules. Algebra, Theorem labelled algebra-theorem-universally-exact-criteria, cited at proof lines 7186-7188; More on Algebra, Lemma labelled more-algebra-lemma-generalized-valuation-ring-modules, cited at proof lines 7188-7191. Reduces universal injectivity to injectivity after tensoring with A/(a).
- Content ideals over a valuation ring. Same chapter, Lemma labelled lemma-flat-finite-type-local-valuation-ring-has-content, cited at proof lines 7195-7197; More on Algebra, Lemmas labelled more-algebra-lemma-content-finitely-generated and more-algebra-lemma-equal-content and Definition labelled more-algebra-definition-content-ideal, cited at proof lines 7197-7203. Produces the principal content ideal of x, identifies it with the content ideal of u(x), and compares it with (a).

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0AT0: sources/stacks-flat/pinned-original-source.tex, source lines 7017–7017. Same-chapter pointer into the author environment situation at source line 7017; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0539: sources/stacks-chow/tags.txt, source lines 6597–6597. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7166 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 053E: sources/stacks-chow/tags.txt, source lines 6602–6602. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7170 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0519: sources/stacks-chow/tags.txt, source lines 6527–6527. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7172 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 00HL: sources/stacks-chow/tags.txt, source lines 647–647. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7175 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7178 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 058K: sources/stacks-chow/tags.txt, source lines 6783–6783. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7184 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ASP: sources/stacks-chow/tags.txt, source lines 13587–13587. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7188 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ASX: sources/stacks-flat/pinned-original-source.tex, source lines 5801–5801. Same-chapter pointer into the author environment lemma at source line 5801; invoked at proof line 7192 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ASB: sources/stacks-chow/tags.txt, source lines 13574–13574. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7195 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 090Q: sources/stacks-chow/tags.txt, source lines 11418–11418. Cross-chapter standard prerequisite located in the project tag list

staged in the local snapshot; invoked at proof line 7197 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ASC: sources/stacks-chow/tags.txt, source lines 13575–13575. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7198 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 0ASA: sources/stacks-chow/tags.txt, source lines 13573–13573. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7201 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *If in Situation 0AT0 the ring A is a valuation ring then the lemma holds.*

Proof. Recall that an A-module is flat if and only if it is torsion free, see More on Algebra, Lemma 0539. Let $T \subset N$ be the A-torsion. Then $u(T) = 0$ and N/T is A-flat. Hence N/T is finitely presented over B, see More on Algebra, Lemma 053E. Thus T is a finite B-module, see Algebra, Lemma 0519. Since N/T is A-flat we see that $T/\mathfrak{m}_A T \subset N/\mathfrak{m}_A N$, see Algebra, Lemma 00HL. As $\overline{u}$ is injective but $u(T) = 0$, we conclude that $T/\mathfrak{m}_A T = 0$. Hence $T = 0$ by Nakayama's lemma, see Algebra, Lemma 00DV. At this point we have proved two out of the three assertions (N is A-flat and of finite presentation over B) and what is left is to show that u is universally injective.

By Algebra, Theorem 058K it suffices to show that $N \otimes_A Q \to M \otimes_A Q$ is injective for every finitely presented A-module Q. By More on Algebra, Lemma 0ASP we may assume $Q = A/(a)$ with $a \in \mathfrak{m}_A$ nonzero. Thus it suffices to show that $N/aN \to M/aM$ is injective. Let $x \in N$ with $u(x) \in aM$. By Lemma 0ASX we know that x has a content ideal $I \subset A$. Since I is finitely generated (More on Algebra, Lemma 0ASB) and A is a valuation ring, we have $I = (b)$ for some b (by Algebra, Lemma 090Q). By More on Algebra, Lemma 0ASC the element $u(x)$ has content ideal I as well. Since $u(x) \in aM$ we see that $(b) \subset (a)$ by More on Algebra, Definition 0ASA. Since $x \in bN$ we conclude $x \in aN$ as desired. □